International University of Sarajevo · EE418 · Final project, 2026

Detecting Illicit Bitcoin Transactions

Graphs, imbalance and drift on the Elliptic dataset

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github.com/berbicanes/EllipticML

The question, and the short answer

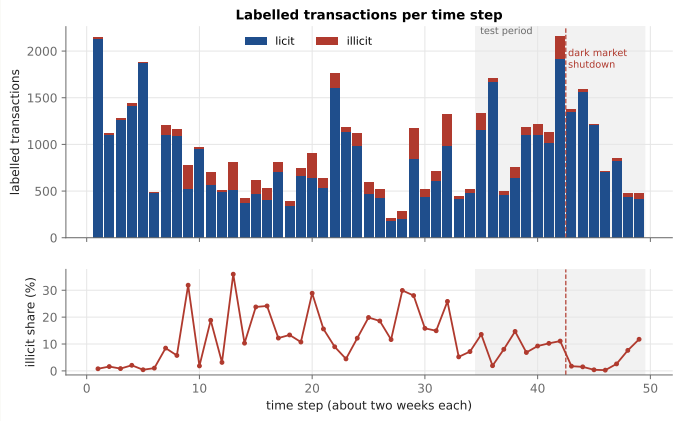
0.804
illicit F1 on unseen future weeks (Random Forest)

+0.06
F1 from neighbour features: the graph helps, mostly through its simplest summary

0.043
F1 after the dark market shutdown: every model fails

— *Detection works, until criminals change behaviour. Most of this talk is about measuring that honestly.*

The data moves under our feet



- 203,769 transactions, 49 two-week steps, 234,355 flows
- Illicit share: 11.6% train, 6.5% test, ≈0 in steps 43–46
- Every step is a separate graph (49 components)
- Illicit neighbours of illicit: 54.1% vs 2.4% for licit
- Illicit transactions never have more than 3 outputs

How we avoided fooling ourselves

Risk	What we did
Training on the future	Paper's temporal split: steps 1–34 train, 35–49 test
Tuning on the test set	Forward folds inside steps 1–34 pick hyper-parameters *and* threshold
A disguised clock	Time step dropped as a feature (keeping it cut LR's F1 from 0.51 to 0.36)
Leaking through preprocessing	Scaling and SMOTE inside the pipeline; graph features label-free
Lucky seeds	Every model $\times$ 5 seeds
False precision	95% CIs by resampling *time steps*, paired tests between models
Silent bugs	Unit tests: no test or future step can reach training

First step: reproduce the paper. Our Random Forest: 0.810 ± 0.005 vs 0.788 published.

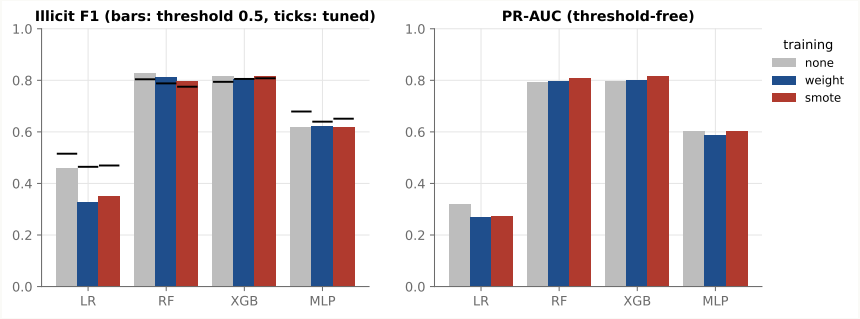
Model comparison: two winners, wide error bars

	F1	95% CI	PR-AUC
Random Forest	**0.804**	[0.618, 0.888]	0.793
XGBoost	**0.794**	[0.613, 0.886]	**0.798**
MLP	0.679	[0.481, 0.750]	0.605
Logistic Reg.	0.515	[0.328, 0.626]	0.319

- RF vs XGBoost: paired test $p = 0.50$ $\rightarrow$ **tied**
- Both beat MLP and LR, $p < 0.001$
- An analyst reviewing the top 100 per step finds 62% of all illicit

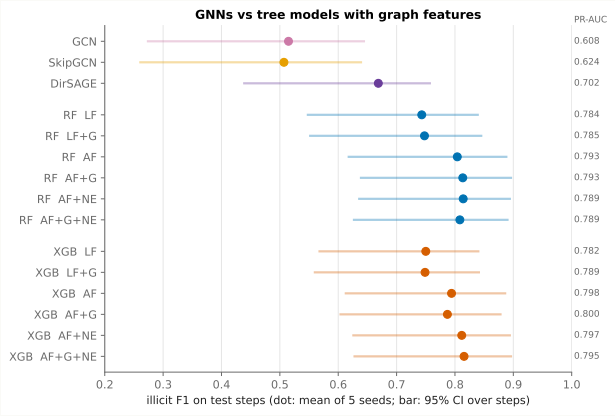
The interval is 0.27 wide: the test set is 15 time steps, not 16,670 independent rows. Many published differences on this benchmark are smaller.

Imbalance: re-balancing barely matters



SMOTE and class weights move PR-AUC by about one point. Threshold tuning helps weak models (LR 0.461 → 0.515) but not the trees: the best threshold drifts over time too.

Does the graph help?



- **Neighbour features: yes.** Local → local + aggregated: 0.743 → 0.804 (p <0.001)
- **More graph: barely.** Structure features and GNN embeddings: $\leq$ 0.02 F1, no PR-AUC
- **GNNs alone: no,** but direction matters: our directed DirSAGE 0.669 vs GCN 0.515

A better GNN makes a worse hybrid

Weber et al. feed GCN embeddings to a Random Forest. We tried the embeddings of our stronger DirSAGE instead.

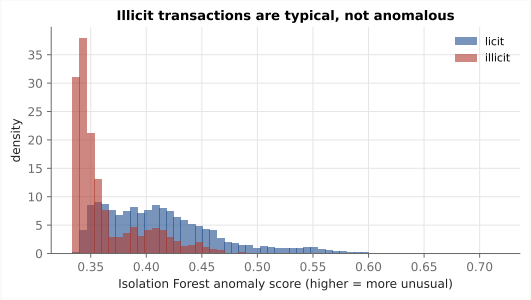
−0.127

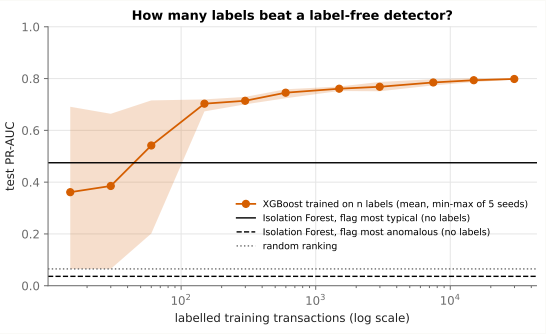
F1 for the Random Forest (p <0.001)

	GCN	DirSAGE
PR-AUC, train rows	0.9797	**0.9999**
PR-AUC, test rows	0.584	0.704
RF weight on embeddings	14.0%	**99.8%**

> *The GNN memorised the training labels, the forest copied it. The published hybrid works because its GNN is weak; embeddings must be made out-of-fold.*

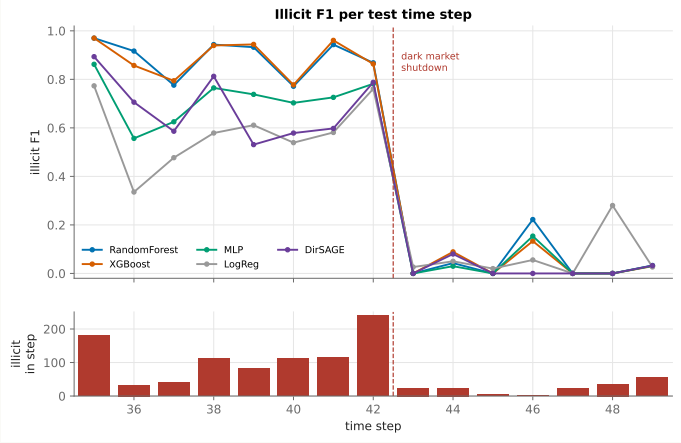
Without labels: crime is typical, not anomalous





- *Flag the most anomalous: PR-AUC 0.036 (worse than random). Flag the most typical: 0.475, no labels. XGBoost needs about 60 labels to beat that.*

Every model fails after the dark market shutdown

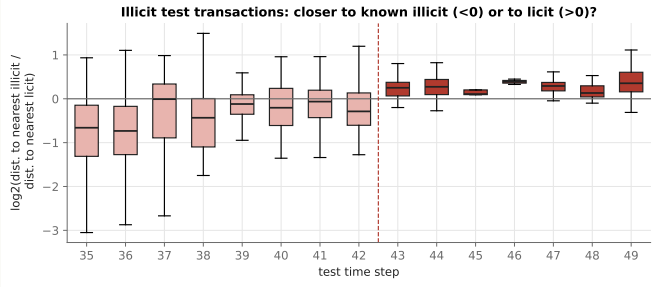


Mean F1 per step, Random Forest:

steps 35–42	**0.890**
steps 43–49	**0.043**

The pooled F1 of ≈0.8 averages two regimes: near-perfect detection of known crime, and almost none of what replaced it.

Why: the new crime looks licit



Closer to licit than to known illicit: 31% before, **90%** after

Features shifted (KS $>$ 0.3): 72% for illicit, 7% for licit

Retraining every step: post-shutdown F1 0.043 $\rightarrow$ 0.185

— *The bottleneck is labels for the new behaviour, not the model. (5 hidden "clock" features were removed first.)*

What I would tell a compliance team

1. Use a tree ensemble on the transaction's own and its neighbours' features.
2. Report uncertainty over *time*, not over rows; test differences in pairs.
3. Do not assume crime is anomalous: here it is the most ordinary traffic.
4. Plan for regime changes: watch the score distribution, label new cases fast.
5. Check stacked models for leakage: a strong upstream model can poison the downstream one.

Limitations: anonymised features; only 169 illicit test transactions after step 43; small GNN search; labels come from heuristics.

Thank you

Questions are welcome.

Code, results, report: github.com/berbicanes/EllipticML

`make all` rebuilds every number from the raw data; every figure in this deck is generated by the pipeline.

Weber et al. (2019), Anti-Money Laundering in Bitcoin, KDD Workshop, arXiv:1908.02591 · Kipf & Welling (2017), ICLR · Hamilton et al. (2017), NeurIPS · Rossi et al. (2023), LoG · Liu et al. (2008), ICDM · Lundberg & Lee (2017), NeurIPS